



Publicly available datasets analysis and spectrogram-ResNet41 based improved features extraction for audio spoof attack detection

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Abstract The rapid expansion of voice-based technologies across diverse applications underscores the critical need for robust security measures against audio spoofing attacks. This paper comprehensively examines publicly available datasets that have been developed to detect audio spoof attacks. The research encompasses a compilation of datasets, including ASVspoof dataset series (2019, 2021), Voice Spoofing Detection Corpus (VSDC), Voice Impersonation Corpus in Hindi Language (VIHL) and DEepfake CROss-lingual evaluation dataset (DECRO), covering various spoofing attack scenarios of English, Hindi and Chinese languages. In the first part of the paper, a baseline for the proposed research work has been developed by comparing the performances of state-of-the-art baseline Linear frequency cepstral coefficient (LFCC) features with application of four different machine learning classifiers Random forest (RF), K-nearest neighbor (KNN), eXtreme gradient boosting (XGBoost), and Naïve Bayes (NB) at the backend, over these four different datasets. In second part of the proposal, we have used novel feature combination of Mel Spectrogram-Residual Network41 (ResNet41)-Linear discriminant analysis (LDA) and Gammatone Spectrogram-ResNet41-LDA, one by one, with application of same set of machine learning classifiers at the backend. The combination of Gammatone spectrogram-ResNet41-LDA along with XGBoost classifier has achieved an Equal Error Rate (EER) of 1.7, 1.28, 0.5, 0.36, 0.03, 0.07, and 0.9% for ASVspoof

2019 Logical Access (LA), ASVspoof 2019 Physical Access (PA), ASVspoof 2021 Deepfake, VSDC, DECRO English, DECRO Chinese, and VIHL datasets, respectively. Hence, the proposed research work in this paper achieves the objective of assessing the feasibility and utility of publicly available state of the art datasets for training and testing advanced algorithms in identifying manipulated audio.

Keywords ASV · Deepfake · Synthetic attack · Replay attack · ResNet41 · Spectrogram

1 Introduction

The proliferation of voice-based technologies across a diverse spectrum of applications has ushered in a new era of convenience and efficiency. However, with this rapid integration comes an urgent need for robust security measures to counteract the rising threat of audio spoofing attacks. These deceptive practices, which involve the imitation or manipulation of audio data, have the potential to compromise the integrity of voice-based systems and applications. This paper addresses this critical concern by conducting a comprehensive assessment of publicly available datasets tailored specifically for the purpose of detecting audio spoof attacks (Dua et al. 2023). The primary objective of this research endeavour is to evaluate the suitability and efficacy of these datasets in training and testing state-of-the-art algorithms designed to identify counterfeit audio.

To achieve this goal, we embark on an exhaustive compilation of diverse publicly accessible datasets, encompassing a wide array of spoofing scenarios including text-to-speech synthesis, voice conversion, and replay assaults. Notably, our dataset collection spans three distinct languages: English, Hindi, and Chinese, providing a comprehensive foundation

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for cross-lingual analysis and evaluation. Each dataset undergoes rigorous scrutiny, with a keen focus on evaluating the diversity of its content, its overall size, and the quality of its annotations. Additionally, we delve into an examination of the representation of various auditory and spectral properties within these datasets, exploring how they can be harnessed to construct robust spoof detection algorithms.

In addressing the verification challenge within speech research, investigations predominantly unfold along two primary trajectories. The initial focus centers on the front-end feature extraction technique, while the subsequent emphasis lies on the back-end classifier tasked with distinguishing between authentic and synthetic audio. Researchers have employed diverse methodologies, with some leveraging conventional feature extraction techniques such as Mel Frequency Cepstral Coefficient (MFCC) (Hossan et al. 2010), Perceptual Linear Prediction (PLP), and Linear Frequency Cepstrum Coefficients (LFCC) (Wang et al. 2019). Concurrently, others have embraced advanced approaches like Constant Q Cepstrum coefficients (CQCC) (Mittal and Dua 2021b), Gammatone Cepstral Coefficients (GTCC) (Valero and Alías 2012), and Basilar-membrane Frequency-band Cepstral Coefficients (BFCC) (Aggarwal and Dave 2011) to derive salient information. Each method exhibits inherent strengths and limitations.

In tandem with advancements in front-end feature extraction, the field has witnessed notable progress in backend classification methodologies. Contemporary practices involve the application of diverse machine learning algorithms for this purpose. For example, Malik et al. (Malik et al. 2020) employed K-nearest neighbour (KNN) (Dhamyal et al. 2021), Naïve Bayes (NB) (Yuslee and Abdullah 2021), and Support vector machines (SVM) (Dhanalakshmi et al. 2009) to construct a binary classifier. Furthermore, deep learning techniques, including Convolutional neural network (CNN) (Mittal and Dua 2021a), Long short-term memory (LSTM) (Joshi and Dua 2022), and their enhanced iterations, play a pivotal role in crafting efficient classification models for Automatic Speaker Verification (ASV) systems.

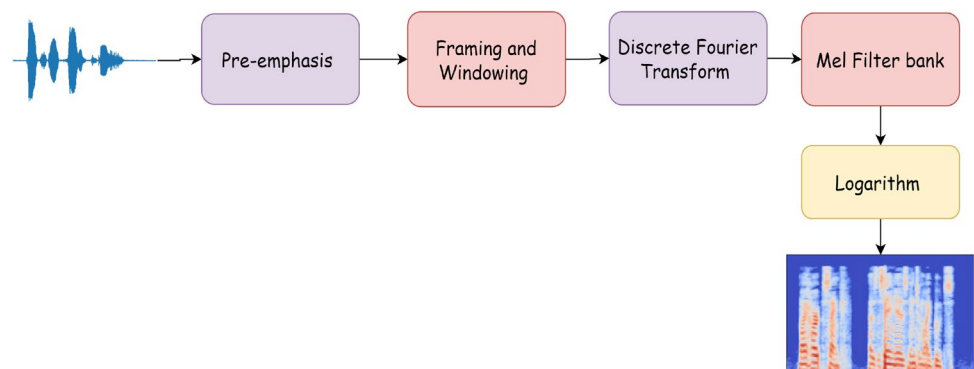
To conduct a thorough performance analysis, we employ a meticulously structured three-phase approach. Firstly, we combine image representation of recorded audio signals Mel Spectrogram (Melspect) and Gammatone Spectrogram (Gammaspect), one by one, with an improved ResNet41 architecture for feature extraction at the frontend. In the second phase, we leverage dimensionality reduction algorithms to discern and select the most discriminative features from the extensive set of 1024 features returned by ResNet41. In the final phase, we leverage widely adopted machine learning techniques, including XGBoost, Random Forest, Naive Bayes, and k-Nearest Neighbors, individually, to benchmark the performance of proposed front end across the chosen datasets.

The remaining sections of the paper are organized as follows: Sect. 1.1 reviews the existing works relevant to the proposed work. Section 2 describes the techniques used to implement the proposed approach. Section 3 details the proposed methodology. Section 4 presents the experiments conducted and results obtained from these experiments, and Sect. 5 discusses the results and provides a comparative analysis of existing methodologies. Finally, Sect. 6 concludes the paper.

1.1 Related work

This section gives the overview of the existing techniques relevant to the proposed work for audio spoof detection. In (Paul et al. 2015), Paul et al. explored eight cepstral features and their corresponding inverted versions. The backend of their proposed approach employed Gaussian Mixture Model (GMM). Remarkably, the double derivative Mel Overlapped Block Transform (MOBT) features outperformed other cepstral features, achieving an impressive average EER of 0.098%, also authors have evaluated their proposed models using ASVspoof 2015 corpora. CQCC, a new feature developed by Todisco et al. (Todisco et al. 2016) have used in their paper to capture detailed features that conventional feature extraction techniques in spoofing detection frequently miss. The authors combined CQCC with a GMM classifier

Fig. 1 Mel Spectrogram Generation



to detect known and new threats using ASVspoof 2015 corpora. Surprisingly, the authors observed a significant relative improvement of 72% over the baseline results in their investigation. In (Chakravarty and Dua 2022), author proposed a noise-robust technique for audio spoof detection. The proposed model is tested in both noisy and clean environments. The frontend of the proposed model is implemented using MFCC, GTCC and fused MFCC-GTCC (Chakravarty Mohit 2023). For classification, TDNN model has been used. The noisy audios have been created by adding noisy samples at 0, 5, and 10 dB from Noizeus database. Fused MFCC-GTCC feature set achieved Min t-DCF of 0.0012, 93% accuracy, 93% precision, 93.5% recall, 93% and F1-score for detecting synthetic attack using ASVspoof 2019 dataset. In (Chakravarty and Dua 2024c), authors have used three spectrograms namely ATP spectrogram, Mel spectrogram, GTCC and MFCC spectrograms for feature extraction and for backend classification pretrained ResNet50 has been used. The proposed model has been evaluated for detecting synthetic, replay attack using ASVspoof 2019 and also a Hindi dataset has been introduced for detecting mimicry attack.

Himavathi et al. (Voice conversion spoofing detection by exploring artifacts estimates 2021) proposed Mel spectrogram, Cochleagram and a classifier based on CNNs to discriminate between the artifact estimates of real speech and voice-converted fake speech in noisy scenarios. This noisy database contained car, train, exhibition, street, and babble noise at 0, 5, 10 and 15 dB. The proposed model has been evaluated using Voice Conversion Challenge 2016 and 2018 datasets and achieved an EER in case of Mel spectrogram-CNN varied from 1.5 to 12.38%, and using Cochleagram-CNN EER varied from 1.87 to 8.50%. Chakravarty et al. (Chakravarty & Dua 2023) used two types of features: image-based features such as ATP and LBP, as well as cepstral features, including MFCC, CQCC, GTCC, BFCC, and PLP features. The backend acoustic models used in the paper included NB, SVM, DT, K-NN, and LSTM (Dua et al. 2022). During the experiment, authors created

fifteen different feature sets by combining ATP, LBP and fused ATP-LBP with all the five cepstral features, respectively. Combining ATP-LBP with GTCC outperformed other sub-models using the LSTM classifier and achieved 99.5% accuracy, 99% precision, 99.2 recall, 98% F1 score, and 0.5% EER for detecting synthetic attack using ASVspoof 2019 LA partition.

In (Ma et al. 2021), authors proposed a new anti-spoofing model named ResWavegram-Resnet (RW-Resnet). The Conv1D ResBlock and backbone Resnet34 are the two components of the proposed model. The collected features are given to Resnet34 in the second step for classification. The proposed model has been evaluated on the ASVspoof 2019 logical access (LA) corpus yielding results of 0.082 min t-DCF and 2.98% EER. Aravind et al. (Aravind et al. 2020) suggested a deep-convolutional neural network-based voice classifier to identify spoofing attempts in ASV systems. Mel-spectrograms and the ResNet-34 architecture were employed in the suggested technique. For the LA scenario, the system obtained an EER of 0.9056%. For the PA scenario of ASVspoof 2019, it achieved an EER of 5.87% on both the development and evaluation datasets. In (Meriem et al. 2023), authors introduced an approach that uses edge detection techniques (Prewitt, Roberts, Sobel, log, Canny) applied to the spectrogram image to capture differences between genuine and spoof speech. SVM has been used as a classifier for implementing the backend of the proposed work. The proposed model has been evaluated using the ASVspoof 2019 dataset and achieved a relative EER improvement of up to 27.62% in the Logical Access scenario and 29.15% in the Physical Access scenario compared to the baseline CQCC feature.

Volkova et al. (Volkova et al. 2019) adopted a deep learning technique based on the Light CNN architecture and applied angular margin-based softmax activation to train a robust deep Light CNN classifier. Fast Fourier Transform (FFT)-spectrograms are taken into account as input by the LCNN. On the assessment portion of the ASVspoof 2017 Version 2 database, the suggested system attained an EER

Fig. 2 Gammatone Spectrogram Generation

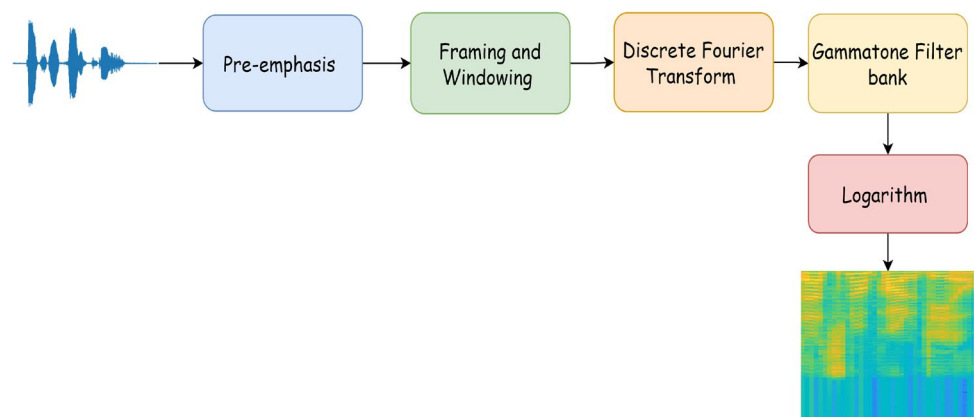


Table 1 Original ResNet50 and improved ResNet41 architecture

Layer name	Output layer	ResNet50	ResNet41
<i>Con_1</i>	112X112	7X7, 64 stride 2	7X7, 64 stride 2
<i>Conv_2</i>	56X56	3X3 MaxPooling, stride 2	3X3 MaxPooling, stride 2
		$\begin{bmatrix} 1,1,64 \\ 3,3,64 \\ 1,1,256 \end{bmatrix} * 3$	$\begin{bmatrix} 1,1,64 \\ 3,3,64 \\ 1,1,256 \end{bmatrix} * 3$
<i>Conv_3</i>	28X28	$\begin{bmatrix} 1,1,128 \\ 3,3,128 \\ 1,1,512 \end{bmatrix} * 4$	$\begin{bmatrix} 1,1,128 \\ 3,3,128 \\ 1,1,512 \end{bmatrix} * 4$
<i>Conv_4</i>	14X14	$\begin{bmatrix} 1,1,256 \\ 3,3,256 \\ 3,3,1024 \end{bmatrix} * 6$	$\begin{bmatrix} 1,1,256 \\ 3,3,256 \\ 3,3,1024 \end{bmatrix} * 6$
<i>Conv_5</i>	7X7	$\begin{bmatrix} 1,1,512 \\ 3,3,512 \\ 3,3,2048 \end{bmatrix} * 3$	–
	1X1	Average pool	Global Pooling layer
		Fully connected Layer	–

of 5.5%. In (Chakravarty & Dua 2024a), the authors used combination of Gammatone spectrograms and ResNet50 for feature extractor for deepfake detection using ASVspoof 2021 dataset and achieved minimum EER 0.4% and an accuracy of 99.7%. Mathew et al. (Mathew et al. 2024) proposed two static deepfake detection models using the ResNet and LCNN architectures. These models have been trained on the ASVspoof 2019 challenge dataset. Various audio feature representations, including Mel-frequency cepstral coefficients (MFCC), power-spectrograms, and Mel-spectrograms, have been empirically tested in the development of proposed deepfake audio model. In the end, Mel-spectrogram features has been selected for the LA model, while power-spectrogram features have been chosen for the PA model. The results show that the LA model has achieved a minimum EER of 7.39%, while the PA model has demonstrated an EER of 4.38%. In (Gomez-Alanis et al. 2019), authors proposed a system that combines an LCNN with RNN for feature extraction. The system can detect different types of spoofing attacks. The proposed Light Convolutional gated recurrent neural network (LC-GRNN) architecture performs better than baseline systems and other deep feature extractors. The system has been evaluated on the ASVspoof 2015, 2017, and 2019 datasets. LC-GRNN with Probabilistic Linear Discriminant Analysis (PLDA) classifier, for ASVspoof 2015, achieved 0.00% EER for known attack and EER of 0.69% for unknown attack. An EER of 3.26, and 6.08% was achieved for ASVspoof 2017 development set and evaluation set, respectively. For ASVspoof 2019 LA set, min t-DCF 0.0000, 0.1523 and EER of 0.00, 6.28% was achieved for development set and evaluation set, respectively. For PA attack, min t-DCF 0.0203, 0.0614 and

EER of 0.73, 2.23% was achieved for development set and evaluation set, respectively.

The literature survey reveals that researchers employ a variety of datasets when developing countermeasures against spoofing attacks. This diversity in datasets presents a challenge, as it complicates the evaluation and comparison of the effectiveness of different countermeasures. To address this issue, it is crucial to systematically analyse and understand the impact of dataset selection on the detection accuracy and reliability of spoofing countermeasures. The pivotal role of datasets in the accurate detection of various audio attacks is a focal point of our investigation. Also, most of the researchers have used commonly used language such as English and Chinese. Motivated by this, our paper aims to assess the influence of diverse datasets on the efficacy of audio attack detection. The evaluation of our model encompasses a comprehensive analysis across multiple datasets, including ASVspoof 2019 (Wang et al. 2019), ASVspoof 2021 (Delgado et al. 2021), Voice spoofing detection corpus (VSDC) (Baumann et al. 2021), Voice Impersonation Corpus in Hindi language (VIHL) (Chakravarty and Dua 2024b) and Deepfake cross-lingual evaluation dataset (DECRO) (Ba et al. 2023). The utilization of these diverse datasets allows for a robust assessment of the model's performance under various conditions. Also, it can be observed from the literature survey that researchers are using different datasets for creating a countermeasure. Hence, this dataset diversity needs to be tackled for checking the importance of dataset for detecting the spoofing attacks. Also, spectrogram plays an important role in detecting the different audio attacks. Hence, motivated by these facts, we have employed two spectrogram-based features in our proposed method namely

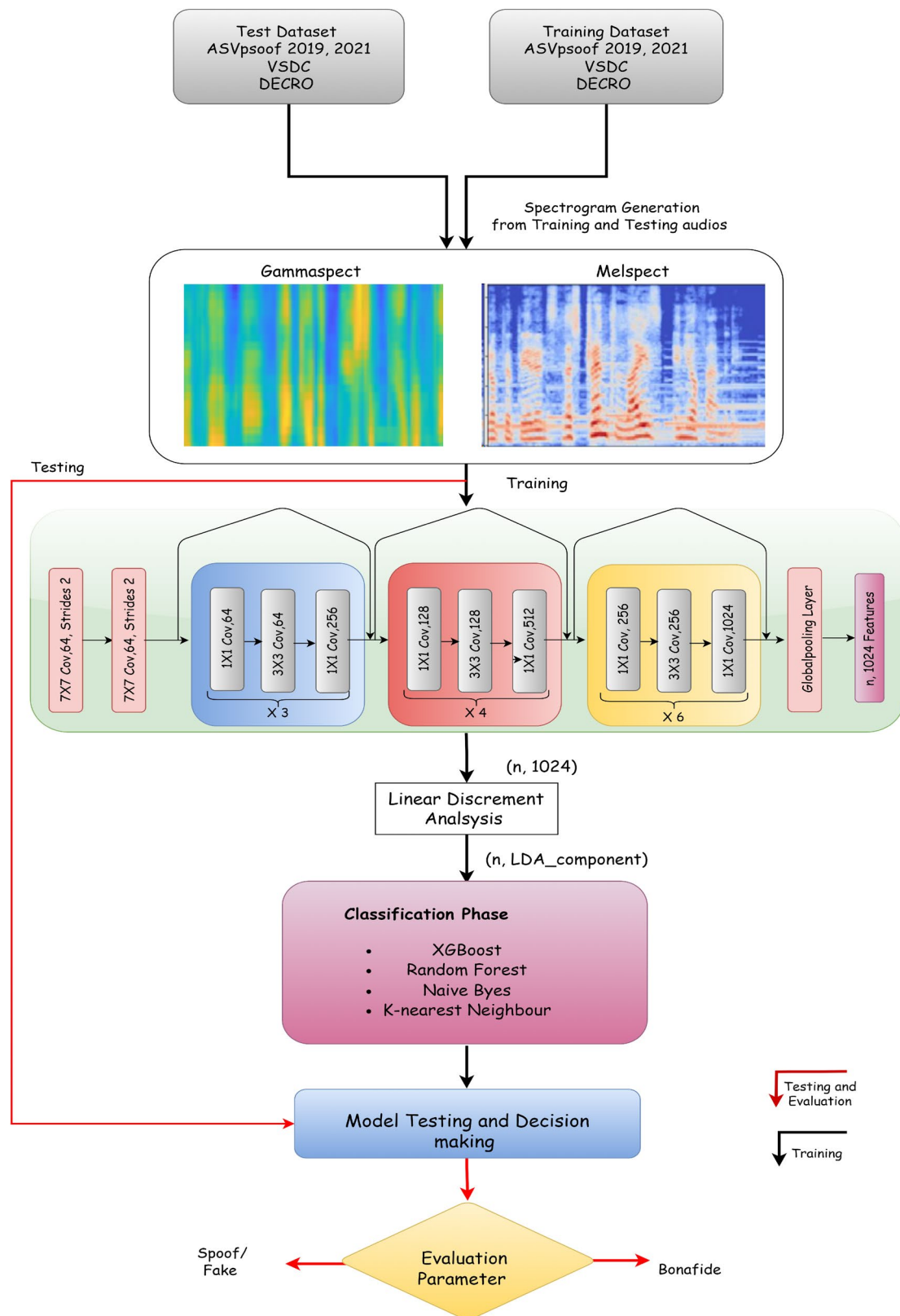
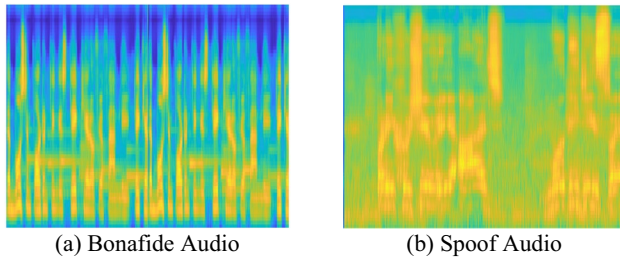
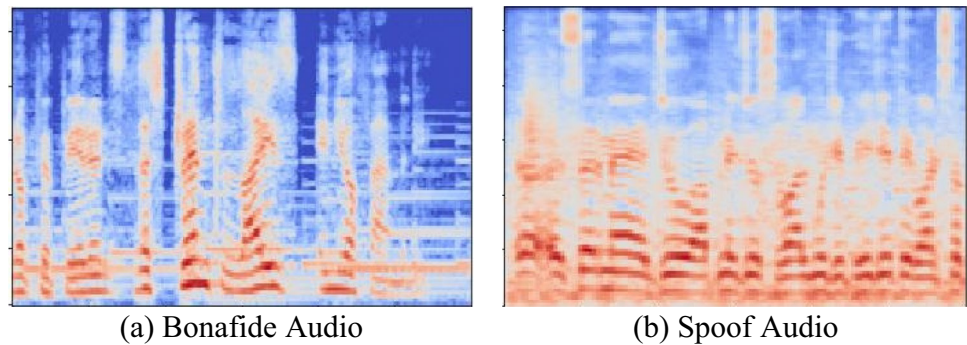


Fig. 3 Flow of Proposed Architecture

Fig. 4 Mel Spectrogram**Fig. 5** Gammatone Spectrogram

Mel Spectrogram and Gammatone Spectrogram. For back-end classification, we have used four different machine learning models such as XGBoost, RF, NB, and KNN. This research has been conducted in two distinct phases. The pointwise contribution of this paper is given below:

1. In the first part of the paper, a baseline for the proposed research work has been developed by comparing the performances of state-of-the-art baseline Linear Frequency Cepstral Coefficient (LFCC) features with application of four different machine learning classifiers Random Forest (RF), K-nearest Neighbor (KNN), eXtreme Gradient Boosting (XGBoost), and Naïve Bayes (NB) at the backend, over four different publically available datasets.
2. In second part of the proposal, we have used novel feature combination of Mel Spectrogram-Residual Network41 (ResNet41)-Linear discriminant analysis (LDA) and Gammatone Spectrogram-ResNet41-LDA, one by one, with application of same set of machine learning classifiers at the backend. Initially, we combine image representation of recorded audio signals Mel Spectrogram and Gammatone Spectrogram, one by one, with an improved ResNet41 architecture for feature extraction at the frontend. Then, we leverage dimensionality reduction algorithms to discern and select the most discriminative features from the extensive set of 1024 features returned by ResNet41. Finally, we leverage four widely adopted machine learning techniques to benchmark the

performance of proposed front end across the chosen datasets.

3. The work evaluates all the proposed models(including baseline) using the ASVspoof dataset2019 & 2021, DEepfake CROss-lingual (DECRO), Voice Spoofing Detection Corpus (VSDC), Voice Impersonation Corpus in Hindi Language (VIHL) by employing performance metrics such as Equal Error Rate (EER) (Cheng and Wang 2004), Accuracy, Precision, F1-score, Recall (Novaković et al. 2017), Detection Error Trade-off (DET) curve(Adler and Schuckers 2005).

2 Preliminaries

This section gives the detailed overview on the techniques that are used to build our proposed model.

2.1 Feature extraction

We have used ResNet41 for feature extraction from spectrograms representation of an audio signal. The spectrogram serves as a graphical representation depicting the strength of a signal in terms of its loudness or intensity, illustrating how the frequencies of acoustic signals change overtime. Typically, the frequency is depicted on the y-axis, time on the x-axis, and the color of the plot indicates intensity. Two types of spectrograms Mel spectrograms (named as Melspect) and Gammatone Spectrogram (named as Gammaspect) to have been used. The following subsections discuss these methods in detail.

- *Mel Spectrogram*: This paper employs the Mel spectrogram, where frequency is presented in the Mel scale. Figure 1 illustrates the process of deriving spectrogram features from auditory signals. This involves dividing the signals into short overlapping windows, transforming them into the frequency domain through Fourier transformation, and then utilizing Mel filter banks to generate the spectrogram envelope. Detailed explanation of spectrogram generation is given below:

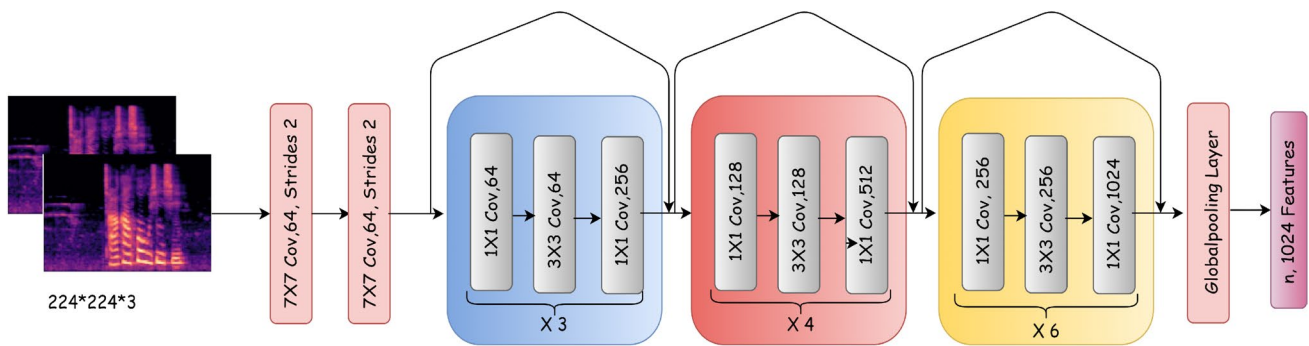


Fig. 6 Architecture of ResNet41

- *Pre-emphasis*: This step applies a filter to the signal to amplify the high-frequency components. The reason for this is to balance the frequency spectrum since higher frequencies tend to have lower amplitudes compared to lower frequencies in speech signals. This helps in improving the signal-to-noise ratio and emphasizes the high-frequency formants.
- *Framing and windowing*: The audio signal is divided into short segments or frames because audio signals are non-stationary, but can be assumed to be stationary in short time intervals.
- *Discrete fourier transform (DFT)*: This step transforms each windowed frame into its frequency components using the Fast Fourier Transform (FFT), which is a computationally efficient way to compute the DFT. This provides the magnitude spectrum for each frame, showing which frequencies are present and their respective amplitudes.
- *Mel filter bank*: The Mel scale is a perceptual scale of pitches judged by listeners to be equal in distance from one another. This step converts the frequency spectrum to a Mel spectrum, which better aligns with human auditory perception.
- *Logarithm*: Human perception of loudness is roughly logarithmic, so applying a logarithm to the Mel spectrum better approximates how humans hear sounds. This step compresses the dynamic range of the Mel spectrum, making it more suitable for further processing and analysis.
- *Spectrogram*: The final output is a spectrogram, which is a visual representation of the audio signal
- *Gammatone spectrogram (Gammaspect)*: Gammatone filters provide a linear approximation of the cochlea's physiologically motivated processing (Holdsworth et al. 1988). It is widely used to represent the human auditory system and is made up of a series of bandpass filters. The first three steps are same as Mel spectrogram generation

as given in above section. The work of last two steps of generating gammatone spectrogram is given below:

- *Gammatone filter bank*: The Gammatone filter bank consists of a set of filters that simulate the frequency selectivity of the human auditory system. When the frequency spectrum is passed through this filter bank, it produces a set of outputs that represent how different frequency components of the signal would be perceived by the human ear. This step transforms the frequency spectrum into a representation that aligns more closely with human auditory perception (Fig. 2).
- *Spectrogram*: The final output is a spectrogram that visualizes how the energy in each Gammatone filter output varies over time.
- *ResNet41*: Inspired by this architecture of ResNet50, we have devised ResNet41 by omitting the last convolutional block of ResNet50 organization. In this improved version ResNet41, the final fully connected layer is substituted with a global pooling layer, specifically tailored for feature extraction. The network generates 1024 units serving as features for respective audio representations. Table 1 gives the description of original ResNet50 and improved ResNet41.

2.2 Dimensionality reduction

We have used Linear Discriminant Analysis (LDA) for reduce the dimensionality of features extracted by ResNet41. LDA is a statistical method used in machine learning and pattern recognition to find the linear combinations of features that best separate two or more classes of objects or events. It works by modelling the difference between classes and seeking to maximize the ratio of between-class variance to within-class variance, thus ensuring that the projected data has the maximum possible separation between different classes. This is achieved through the computation

Table 2 ASVspoof 2019 data set

Dataset Type	Total speakers	# LA Audio observations	# PA Audio observations
Training Set	20 (8 M, 12F)	25,380(2580 Original, 22,800 spoofed)	54,000(5400 Original, 48,600 spoofed)
Development Set	35(8 M, 12F)	24,844(2548 Original, 22,296 Spoofed)	29,700(5400 Original, 24,300 spoofed)
Evaluation Set	46(20 M,26F)	1,43,000(8000 Original, 1,35,000 Spoofed)	1.43,000(8000 Original, 1,35,000 Spoofed)

Table 3 ASVspoof 2021 data set

Dataset type	Total speakers	Number of audio
LA	67 (30 M, 37F)	16,492(bonafide), 148,148(spoof)
PA	67 (30 M, 37F)	108,540(bonafide), 699,840(spoof)
DF	93(43 M,50F)	20,637(bonafide), 572,616(spoof)

of eigenvectors and eigenvalues from the data covariance matrix, which are then used to construct a linear transformation. The application of LDA involves the following steps:

- Determine d -dimensional mean vectors for every class in the dataset.
- Determine scatter matrices, both between and within classes.
- Determine the eigenvalues ($e_1, e_2, \dots, e_d$) and corresponding eigenvectors ($\lambda_1, \lambda_2, \dots, \lambda_d$) for the scatter matrices.
- Arrange eigenvectors in a $d \times k$ – *dimensional* matrix by decreasing eigenvalues. Choose the k eigenvectors with the highest eigenvalues (each column represents an eigenvector).
- Apply the $d \times k$ eigenvector matrix to transform the data into a new subspace. The transformation is represented as $Y = X \times W$, where X is an $n \times d$ -dimensional matrix representing n samples and Y is the converted $n \times k$ -dimensional sample set in the new subspace.

2.3 Classification

This section provides an overview of the classification algorithm employed in constructing the backend infrastructure of our proposed model. We have used widely adopted machine learning techniques RF, NB, XGBoost, and KNN.

- **XGBoost:** XGBoost is a sophisticated parallel tree boosting algorithm intended for large-scale situations requiring high precision (Singh et al. 2022). XGBoost builds an ensemble of decision trees sequentially, where each new tree attempts to correct the errors made by the previous trees. This is done by fitting the new tree to the gradient of the loss function. In (Rahmeni et al. 2022), the authors used the XGBoost classifier for speech spoof

Table 4 VSDC dataset

Sample Type	Audio samples		
	0	1	2
Count of Samples	1687	6179	6184

detection, which improved the speed and performance of the machine learning model.

- **RF:** Random Forest is an ensemble algorithm that employs decision trees as base classifiers (Breiman 2001). The development of a Random Forest involves three primary stages. First, to increase ensemble diversity, Random Forest manipulates training sets using bootstrap sampling to generate multiple learning sets. Next, it uses random trees on these training sets, where a few input features are randomly selected for each node.
- **KNN:** KNN is a supervised learning algorithm that classifies new instances based on their proximity to training examples in the feature space. When test results are submitted, they are assigned to the class with the highest representation among their nearest neighbors. To classify a new instance, the Euclidean distance between it and all previously saved training audio clips is computed (Murugappan 2011). The majority voting method is used to categorize unclassified data.
- **NB:** A Naive Bayes classifier applies Bayes' theorem with the "naive" assumption that all predictors are independent given the class (Fu et al. 2010). To classify a new instance X , the Naive Bayes classifier computes the posterior probability for each class and assigns the class with the highest probability.

3 Proposed work architecture

Figure 3 illustrates the procedural flow of the proposed methodology designed for the detection of various spoofing attacks. Within the framework of our proposed approach, the principal emphasis resides in the examination of how the dataset influences the efficacy of the spoof detection system. We begin by compiling a broad set of publicly available datasets such as ASVspoof 2019 LA & PA, ASVspoof 2021 DF, VSDC, VIHL and DECRO. These datasets are

Table 5 DECRO dataset

Samples	English			Chinese		
	Partition					
	Training	Evaluation	Development	Training	Evaluation	Development
Bonafide	5129	4306	3049	9000	6109	6109
Spoofed	17,412	14,884	10,503	17,850	12,015	12,015

Table 6 VIHL dataset

Partition	Voice impersonation corpus in Hindi language	
Training	Bonafide	Impersonated
Evaluation	150	100
Development	50	50

used for training and testing the model. The feature extraction process entails the generation of two spectrograms, namely Melspect and Gammaspect, from the training set of the aforementioned datasets. For feature extraction from these spectrograms, an improved ResNet41 architecture is employed, resulting in the extraction of 1024 deep features from audio inputs. Acknowledging that not all 1024 features significantly contribute to the detection process, a discerning choice is made to employ Linear Discriminant

Analysis (LDA) to select the most discriminative features. These selected features are subsequently utilized for training classification models. We have employed machine learning techniques, specifically RF, NB, XGBoost, and KNN. These models' performance is rigorously benchmarked across the prescribed datasets, using assessment criteria such as Equal Error Rate (EER), accuracy, precision, recall, F1-score, DET curve and t-DCF to assess their efficacy.

3.1 Pre-processing

This section provided an overview of audio data pre-processing used for creating the datasets. ASVspoof 2019 database is divided into Logical Access (LA) and Physical Access (PA) scenarios, both derived from the VCTK corpus with data from 107 speakers. The LA partition includes bona fide and spoofed speech from 17 TTS and VC systems, with 6 known attacks

Table 7 EER (%) of baseline LFCC features with ML classifiers over different dataset

Frontend Technique	Classification models	ASVspoof 2019		ASVspoof 2021	DECRO		VSDC	VIHL
		LA	PA		English	Chinese		
		Equal Error Rate (%)						
LFCC	XGBoost	15.5	21.1	37.1	2.8	6.06	1.5	5.8
	Random forest	17.1	21.2	39.3	2.3	6.87	4.2	6
	Naïve Byes	16.6	21.2	40.6	13	10.4	26	8.4
	KNN	32.4	32	42.8	2.8	12.3	3.8	8.3

Table 8 Performance of proposed feature combinations using ASVspoof 2019 LA

Frontend techniques	Classification Models	EER (%)	Accuracy	Recall	F1-Score	Precision
Mel spectrogram-ResNet41-LDA	XGBoost	5.00	93	94	99	90
	Random Forest	6.76	91	92	99	86
	Naïve Byes	5.81	92	93	99	83
	KNN	6.02	92	93	99	88
Gammatone spectrogram- ResNet41-LDA	XGBoost	1.76	98	98	98	98
	Random Forest	3.40	96	96	96	96
	Naïve Byes	2.51	97	97	97	97
	KNN	1.88	98	98	98	97

in the training and development sets and 11 unknown attacks in the evaluation set. The PA partition contains bonafide and spoofed speech in a reverberant environment, with various acoustic and replay configurations. ASVspoof 2021 database introduces new partitions for LA, PA, and Deepfake (DF) tasks, requiring participants to use ASVspoof 2019's training and development data. LA evaluation includes trials with real telephony systems' artifacts, while PA evaluation data is recorded in real physical environments. DF evaluation data

features audio with coding and compression artifacts, with all data provided in 16-bit PCM at a 16 kHz sampling rate.

Voice Spoofing Detection Corpus (VSDC) includes genuine and replay recordings at a 96,000 Hz sampling rate, capturing audio with professional microphones and mobile devices. The data is exported as WAV files at 96,000 Hz, ensuring high quality. DECRO consists of English and Chinese subsets, containing bona fide and spoofed speech samples with nearly identical audio lengths. Spoofed samples are generated using various TTS and VC algorithms. The Chinese data comes from

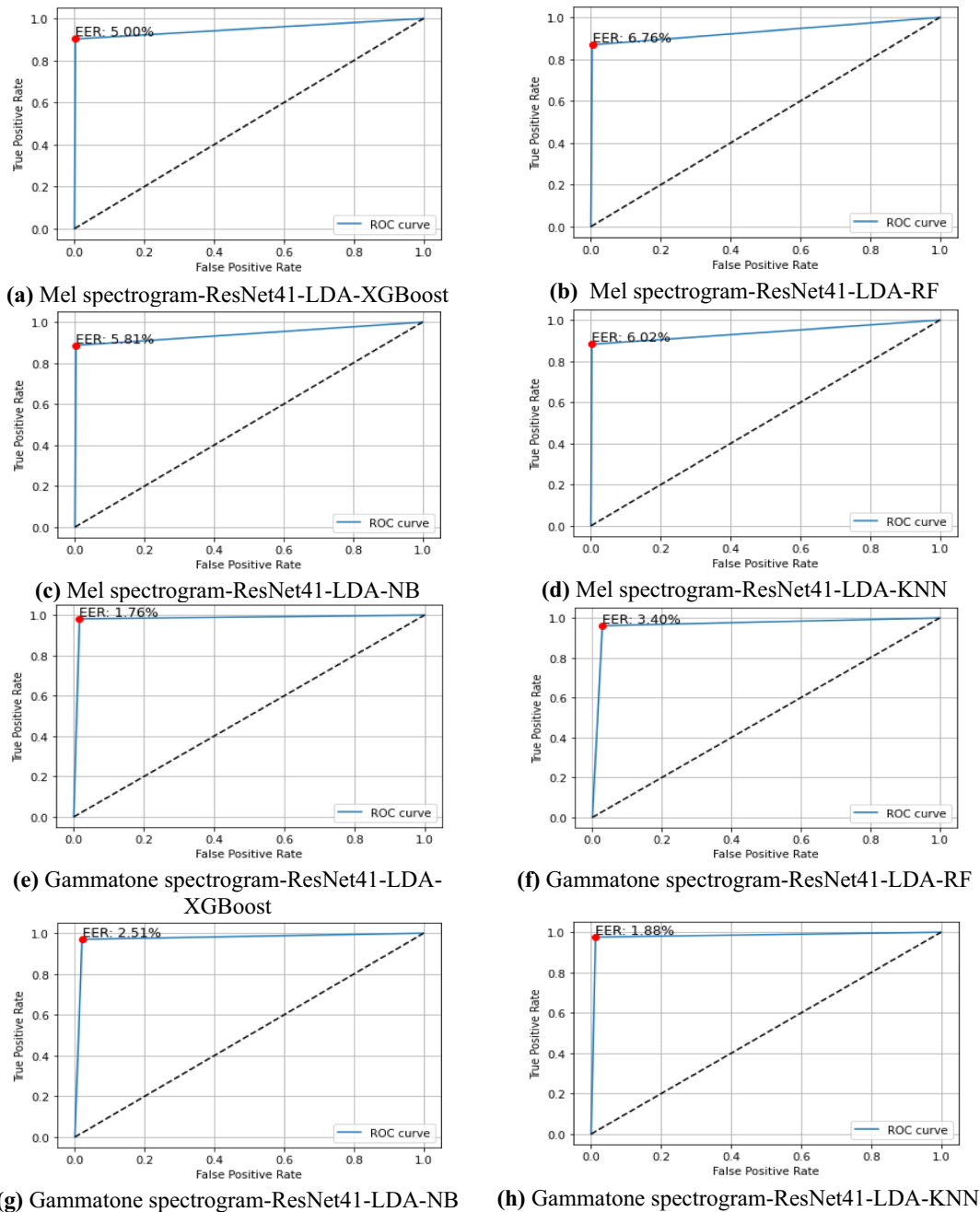


Fig. 7 DET Curve for Proposed Feature Combinations using ASVspoof 2019 LA

Table 9 Performance of proposed feature combinations using ASVspoof 2019 PA

Frontend techniques	Classification Models	EER (%)	Accuracy	Recall	F1-Score	Precision
Mel spectrogram-ResNet41- LDA	XGBoost	12.23	89	84	90	80
	Random Forest	19.13	83	75	78	72
	Naïve Byes	12.44	89	84	90	80
	KNN	15.13	87	80	85	77
Gammatone spectrogram-ResNet41-LDA	XGBoost	1.28	99	99	99	99
	Random Forest	3.40	96	96	96	96
	Naïve Byes	2.51	97	97	97	97
	KNN	5.00	95	95	95	95

the Wavefake dataset, while the English data is sourced from ASVspoof 2019 and other public datasets. The VIHLL dataset includes genuine and Hindi-fake audios from YouTube, with environmental disturbances and impersonated audios created using AI tools or voice mimicry. The audio is segmented into 3-s clips.'

3.2 Feature extraction

This phase plays a pivotal role in identifying audio spoof attacks. The suggested methodology involves converting the audio signal into spectrograms. The spectrograms are then fed into a ResNet41 model, which extracts deep information about audios. This section describes the approaches utilized to implement the front-end of the proposed model.

- *Mel spectrogram*: Function 1 outlines the process of extracting the Mel spectrogram. This spectrogram employs the Mel scale, a perceptual pitch scale designed to approxi-

mate human perception of frequency differences (Huang et al. 2001). This method provides a succinct and meaningful representation of audio signals that aligns closely with human perception. Figure 4 shows output of Function 1 for bonafide and spoofed audio. The *getMelSpect()* is created in MATLAB. This function takes audios as input. The line 2 of Function generates a Melspect from the audio signal *y* using the specified sampling rate *sr*. It uses an FFT window size of 2048, a hop length of 512, and computes 100 Mel bands. Line 3, converts the Mel spectrogram from power scale to decibel (dB) scale, resulting in a log-Mel spectrogram stored in the variable *log_mel_spectrogram*. Line 4 displays the log-Mel spectrogram with time on the x-axis and Mel frequencies on the y-axis using the specified *sr*. Line 5 extracts the base name of the audio file from its full path and stores it in the variable *filename* in specified format.

Function 1: <i>getMelSpect()</i>
Input: <i>path_to_audio</i>
1. <i>y, sr = librosa.load(audiofile_path)</i> //Load the audio file
2. <i>mel_spectrogram = librosa.feature.melspectrogram(y, sr = sr, n_fft = 2048, hop_length = 512, n_mels = 100)</i> //Calculates the Mel spectrogram
3. <i>log_mel_spectrogram = librosa.power_to_db(mel_spectrogram)</i> //Converts the Mel spectrogram to a logarithmic scale
4. <i>librosa.display.specshow(log_mel_spectrogram, x_axis = 'time', y_axis = 'mel', sr = sr)</i> // Display Melspectrogram
5. <i>filename = os.path.basename(audiofile_path)</i>
6. <i>plt.savefig(filename + '.jpg')</i> //Save Spectrograms
Output: Melspect for each audio

Fig. 8 DET Curve for Proposed Feature Combinations using ASVspoof 2019 PA

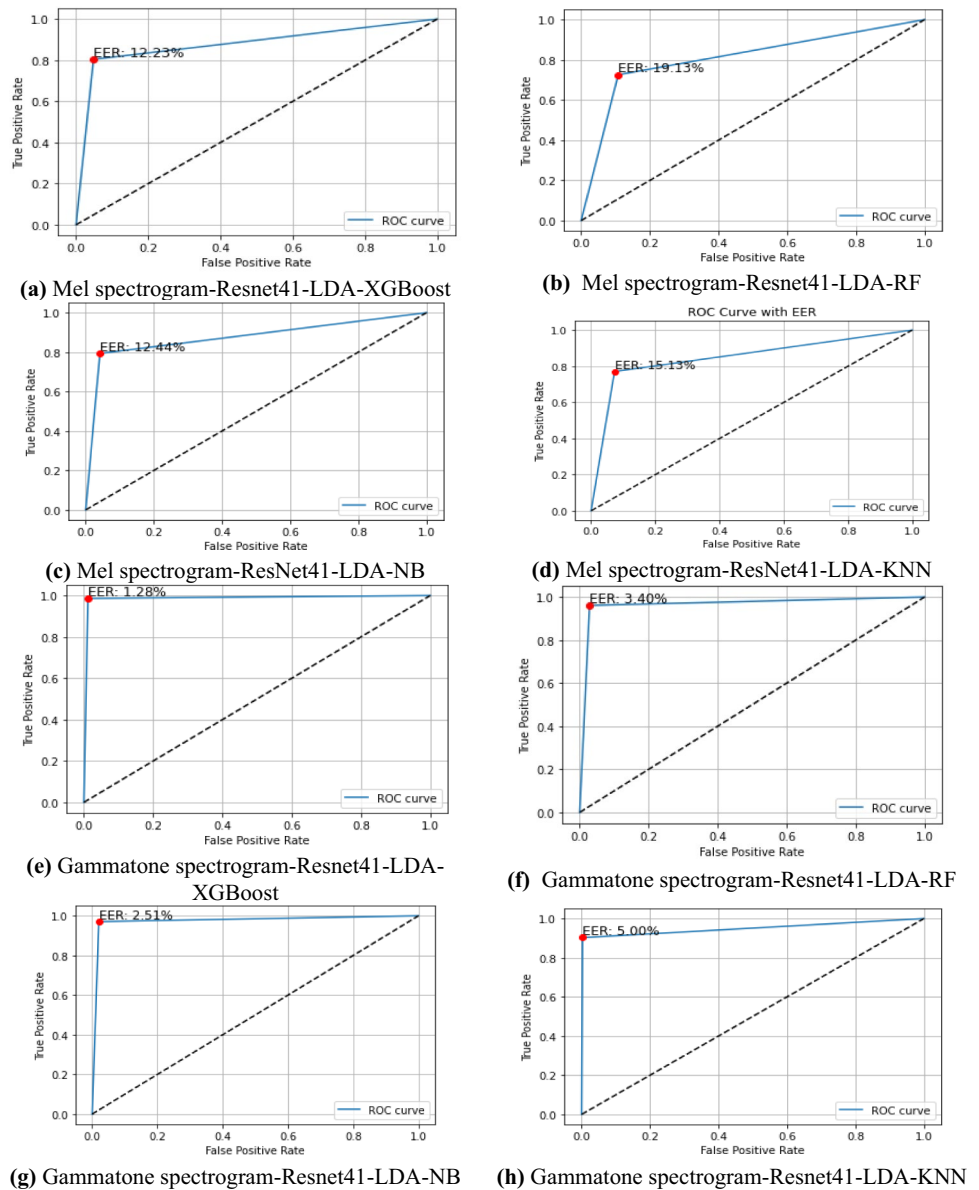


Table 10 Performance of proposed feature combinations using ASVspoof 2021 DF

Frontend techniques	Classification Models	EER (%)	Accuracy	Recall	F1-Score	Precision
Mel spectrogram-ResNet41-LDA	XGBoost	5.00	95	95	95	95
	Random Forest	5.00	95	95	95	95
	Naïve Byes	5.00	95	95	95	95
	KNN	5.00	95	95	95	95
Gammatone spectrogram-ResNet41-LDA	XGBoost	0.55	99	99	100	99
	Random Forest	0.92	98	98	100	98
	Naïve Byes	0.61	98	99	100	98
	KNN	0.80	98	99	100	98

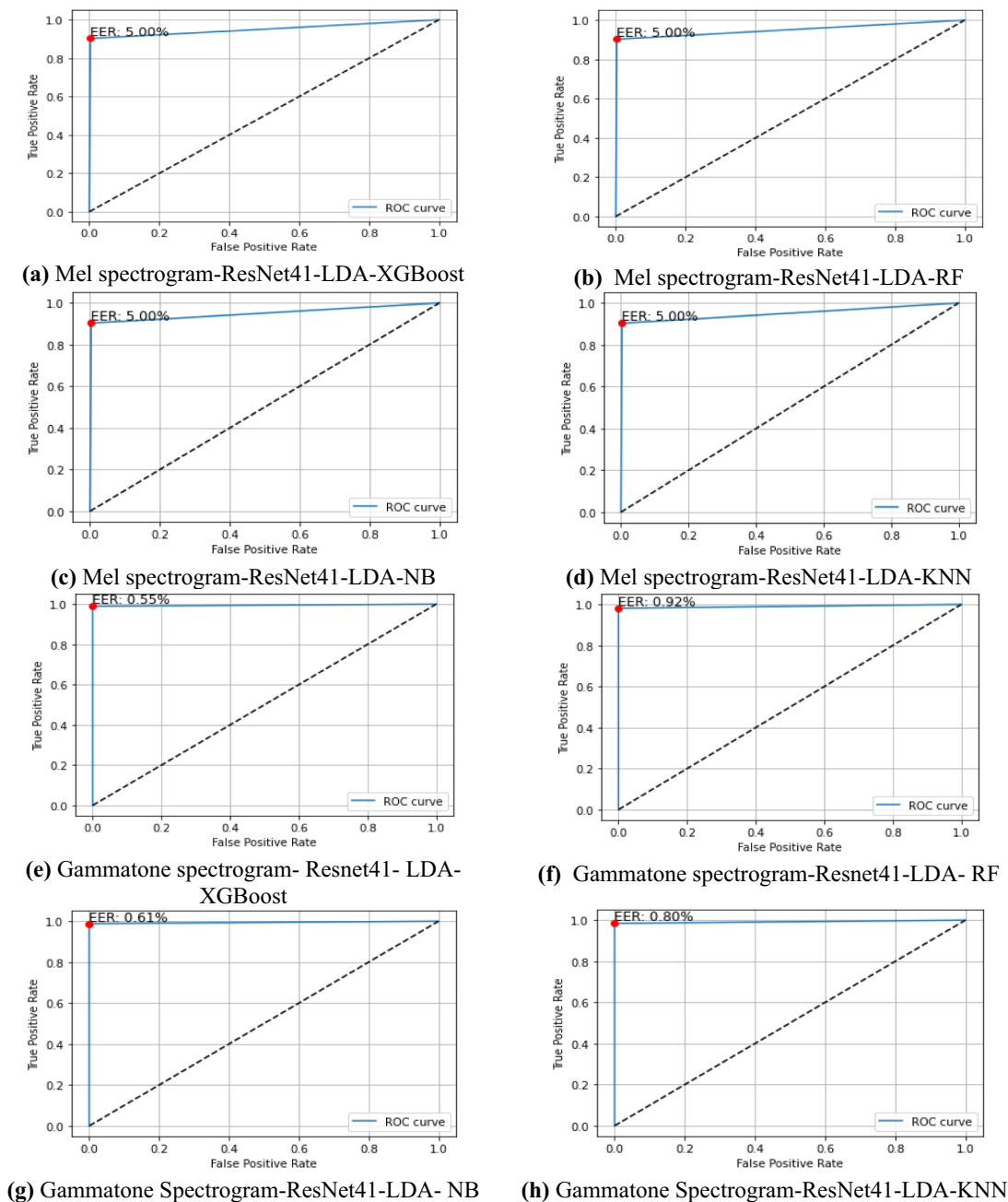


Fig. 9 DET Curve for Proposed Feature Combinations using ASVspoof 2021 DF

- Gamma spectrogram:** This methodology provides a straightforward means of generating time–frequency surfaces through gammatone analysis. While spectrograms are prevalent for visualizing time–frequency relationships, they deviate significantly from the auditory analysis of sound (Slaney 1993). Function 2 details the process of extracting the Gammatone spectrogram. Figure 5 shows gammatone spectrogram generated for

bonafide and spoofed audios generated by Function 2. Function `getGammaSpect()` is created using MATLAB function `gammatonegram`. Line 5 reads the audio file specified by filename1 and stores the audio data in *d* and the sampling rate in *sr*. Line 7 represents a function call where `gammatonegram` is a function that computes and displays a Gammaspect of the audio signal *d* with a sampling rate *sr*.

Function 2: <i>getGammaSpect()</i>
Input: <i>path_to_audio, extension, n,</i> <i>path_to_audios, path_to_save_spectrogram</i>
<pre> 1. <i>audio_files</i> = <i>dir(fullfile(path_to_audio, ['*.', extension]));</i> 2. <i>for</i> <i>i</i> = 1:<i>n</i> 3. <i>filename</i> = <i>audio_files(i).name;</i> 4. <i>filename1</i> = <i>fullfile(path_to_audios, filename);</i> 5. [<i>d, sr</i>] = <i>audioread(filename1);</i> 6. <i>d</i> = <i>reshape(d, [], 1);</i> 7. <i>D</i> = <i>gammatonegram(d, sr);</i> 8. <i>imagesc</i>(20 * <i>log10(D)</i>); 9. <i>saveas(gcf, fullfile(output_path, ['spectrogram', num2str(i), '.png']));</i> </pre>
Output: <i>Gammspect for each audio</i>

- *Improved residual network (ResNet41)*: ResNet41 is a deep Convolutional Neural Network (DCNN) composed of 41 layers, comprising two types of blocks: Conv Block and Identity Block. ResNet50 is well known for its capacity to extract hierarchical and discriminative features from input spectrograms. Inspired by this, we created our deep neural network by deleting ResNet50's final convolutional block. Figure 6 depicts the redesigned architecture of ResNet50, which resulted in ResNet41.

Function 3 show the implementation of improved ResNet41 along with LDA. In Line 1, the ResNet41 model is initialized using the 'imagenet' weights as an argument. In Line 2, the model is configured with an input and an output layer, where the output layer is set to an average pooling layer to extract features from the ResNet41 model. Line 5 indicates the number of features extracted, which are stored in a feature array. In Line 7, the LDA function is applied to the feature array for dimensionality reduction.

Table 11 Performance of proposed feature combinations using VSDC

Frontend techniques	Classification Models	EER (%)	Accuracy	Recall	F1-Score	Precision
Mel spectrogram-ResNet41-LDA	XGBoost	0.56	99	99	99	99
	Random Forest	1.61	98	99	99	98
	Naïve Byes	0.70	98	99	99	98
	KNN	1.28	99	99	99	99
Gammatone spectrogram-ResNet41-LDA	XGBoost	0.36	99	99	99	99
	Random Forest	0.81	99	99	99	99
	Naïve Byes	0.46	99	99	99	99
	KNN	0.61	99	99	99	99

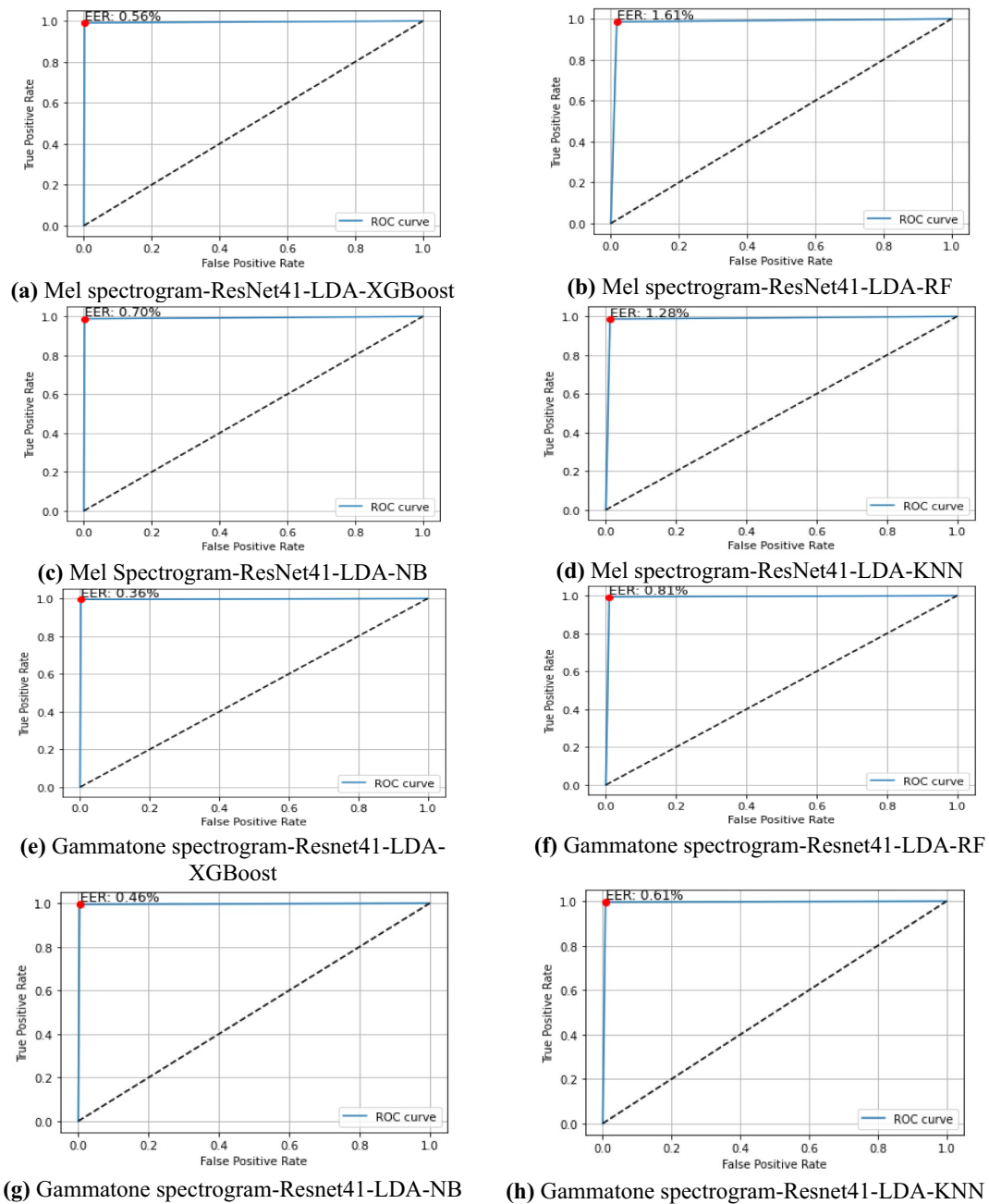


Fig. 10 DET Curve for Proposed Feature Combinations using VSDC

Table 12 Performance of proposed feature combinations using DECRO English

Frontend techniques	Classification Models	EER (%)	Accuracy	Recall	F1-Score	Precision
Mel spectrogram-ResNet41-LDA	XGBoost	1.28	99	99	99	99
	Random Forest	2.51	97	97	97	97
	Naïve Byes	1.88	98	98	98	97
	KNN	5.15	95	95	95	95
Gammatone spectrogram-ResNet41-LDA	XGBoost	0.03	99	99	99	100
	Random Forest	0.06	99	99	99	99
	Naïve Byes	0.05	99	99	99	100
	KNN	0.05	99	99	99	100

Function 3: *ResNeFext(spectrogram)***Input:** *Mel spectrogram or Gammatone Spectreogram*

```

1. base_model = ResNet41(weights = 'imagenet')
2. model = Model(inputs =
   base_model.input, output base_model.get_layer('avg_pool').output)
3. image_size = 224
4. img_paths = img_df.Path.tolist()
5. features_array = np.zeros((n_audios, 1024))
6. num_unique_classes = len(set(y)) // y represents Labels
7. lda = LDA(n_components = min(num_unique_classes - 1, 1))
8. X_train = lda.fit_transform(X_train, y_train)
9. X_test = lda.transform(X_test)

```

Output: *X_train, X_test feature vector*

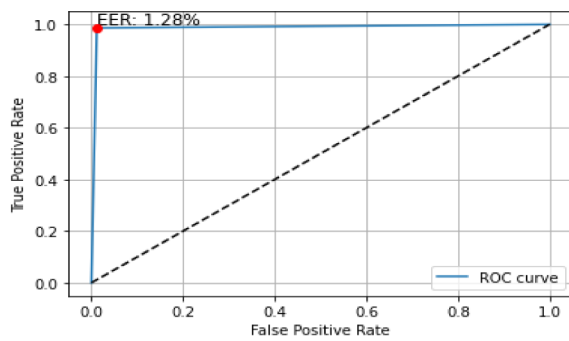
3.3 Backend classification

To implement back-end of our proposed model, four Machine Learning (ML) algorithms NB, RF, XGBoost and K-NN have been used. In our proposed work, we have utilized XGBoost for spoof attack detection. The model was configured with various hyperparameters such as *learningrate* of 0.001, *anensemble* of 1000 *estimators(trees)*, a *maximumtreedepth* of 5, a *minimumchildweight* of 1, a *gamma* value of 0.1, and both data and feature *subsampling* rates of 0.8. Additionally, *L1 regularization* with a strength of 0.005 was incorporated, four processing threads were utilized, a positive class *weightscale* of 1 was established, and a *fixedseed* value of 27 has employed to ensure reproducibility. For performing the experiments, RF has been implemented using *n_estimator*, *max_depth*, *min_(samples_split)*, *min_(samples_leaf)* parameters, where *n_estimator* ranging from {50, 100, 200}, *max_depth* extends from {None, 10, 20}, *min_(samples_split)* ranging from {2, 5, 10},

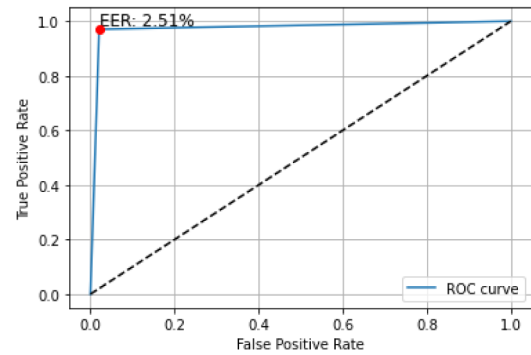
and *min_(samples_leaf)* covers from {1, 2, 4}. Gaussian distribution has been used for NB experiment. The implementation is done using the built functions in the sklearn library. For KNN trials, we carefully tuned two critical parameters: the number of neighbours K_N and the distance metric D_m . We evaluated the performance for various K_N that is (1, 10, 100) and D_m alternatives (Euclidean, Manhattan Distance, Chebyshev Distance). We consistently got optimal results in all three categories using weighted KNN with $K_N=10$ and $D_m=Euclidean$.

4 Experimental analysis and results

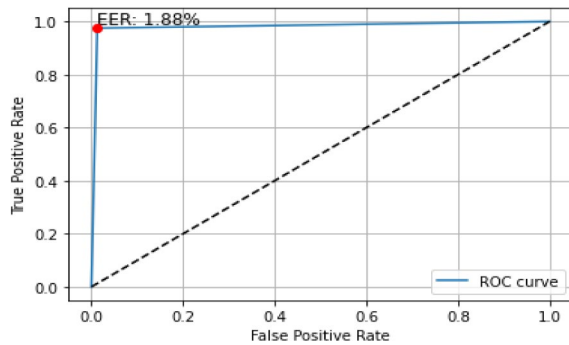
In this section, we offer a detailed insight into the experiments conducted to evaluate the efficacy of our spoof detection methodology. Our approach was implemented using MATLAB 2022 for the front-end, while the back-end model has been coded in Python 3.7 and executed using Anaconda. The computational resources utilized include a computer



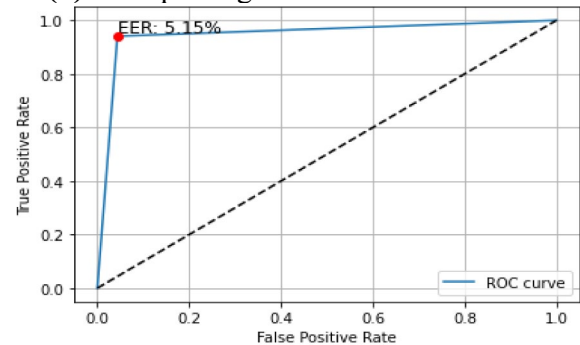
(a) Mel spectrogram-ResNet41-LDA-XGBoost



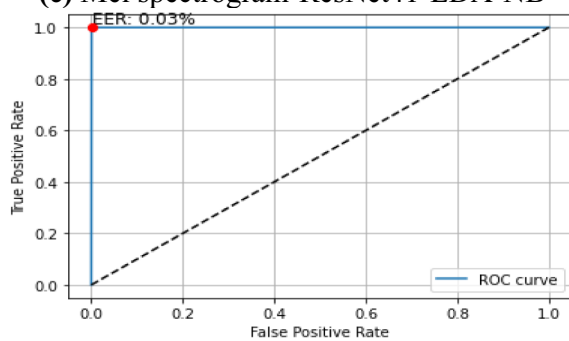
(b) Mel spectrogram-ResNet41-LDA-RF



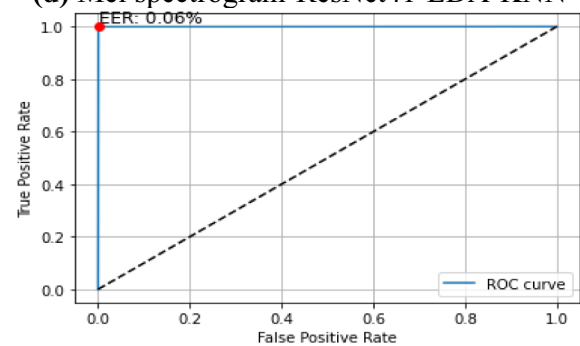
(c) Mel spectrogram-ResNet41-LDA-NB



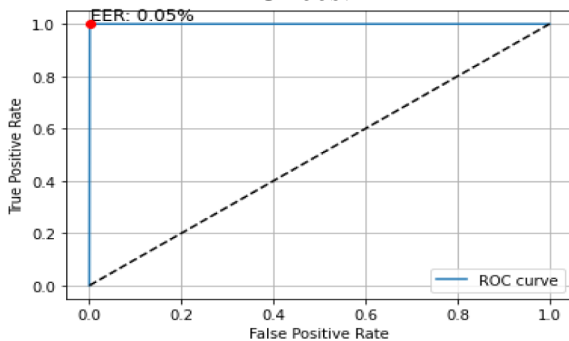
(d) Mel spectrogram-ResNet41-LDA-KNN



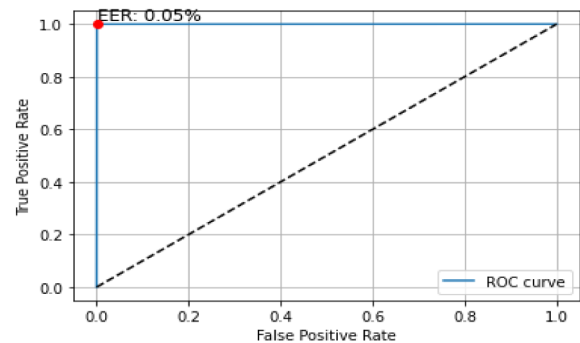
(e) Gammatone spectrogram-Resnet41-LDA-XGBoost



(f) Gammatone spectrogram-Resnet41-LDA - RF



(g) Gammatone spectrogram-Resnet41-LDA -NB



(h) Gammatone spectrogram-Resnet41-LDA -KNN

Fig. 11 DET Curve for Proposed Feature Combinations using DECRO English

Table 13 Performance of proposed feature combinations using DECRO Chinese

Frontend techniques	Classification Models	EER (%)	Accuracy	Recall	F1-Score	Precision
Mel spectrogram-ResNet41-LDA	XGBoost	0.55	99	99	99	99
	Random Forest	0.92	99	99	99	99
	Naïve Byes	0.61	99	99	99	99
	KNN	0.80	99	99	99	99
Gammatone spectrogram-ResNet41-LDA	XGBoost	0.07	99	99	99	99
	Random Forest	0.11	99	99	99	99
	Naïve Byes	0.09	99	99	99	99
	KNN	0.09	99	99	99	99

with an 11th generation Intel Core i5 processor and 16 GB of RAM. We provide a comprehensive description of the datasets employed in our proposed work, followed by the performance analysis of the baseline and proposed ASD models.

4.1 Publicly available datasets

This paper focuses on the databases used to develop and evaluate countermeasure systems (CMs) against spoofing attacks.

- *ASVspoof 2019*: ASVspoof 2019 (Yamagishi et al. 2019) dataset contains Logical Access (LA) partition for synthetic attacks and the Physical Access (PA) partitions for replay attacks. Audio samples from Eight males and Twelve females have been considered in training and development sets. The training and development sets for both LA and PA attacks have been developed using same type of the spoofing algorithms. Several ASVspoof competitions have been held in the last decade, each addressing a distinct scenario and utilising progressively complex approaches. The ASVspoof challenge held in 2019 challenges cutting-edge Text to Speech (TTS) and Voice Conversion (VC) algorithms that have been used to create spoofing attacks for LA and PA detection tasks. However, these datasets are imbalanced which affects the performance of countermeasures. Other limitation of ASVspoof datasets is that these datasets consider only one language that is English. Table 2 gives the detail of the ASVspoof 2019 dataset.
- *ASVspoof 2021*: Recently introduced ASVspoof 2021 dataset (Liu et al. 2022) contains more complex data as compare to previous datasets available. It is considered as the most difficult scenario to date. A new DF detection task has been included where no ASV system was present, making it even more complicated and practical. It focuses on both logical access attacks and replay attacks similar to ASVspoof 2019 dataset, alongwith one more complex class of spoofing attacks called Speech Deepfake. The dataset has been divided into three sets: training set, development set and evaluation set. The data provided in training and development set is labelled, whereas data of evaluation set is unlabelled. Table 3 gives details about ASVspoof2021.
- *VSDC*: VSDC (K. Malik et al. 2020) primarily represents different replay assaults. This dataset consists different folders, each of which contains authentic (bonafide) recordings as well as the original user's first-point replay (1PR) and second-point replay (2PR) assaults. This dataset has been created under controlled environment. Hence, its audio samples do not contain noise. The employment of replay speakers and recording equipment is required to implement 1PR and 2PR assaults (K. M. Malik et al. 2019), (Baumann et al. 2021). Each user gives nine distinct sentences inside each recording. Table 4 represents the samples collected at 0, 1, and 2 replay points.
- *DECRO*: Authors have chosen English and Chinese, the two most widely spoken languages, to build the DECRO dataset to maximize its potential benefit for the research community (Ba et al. 2023). Both the English and Chinese sections feature authentic and spoofed speech samples, with nearly equal total audio lengths. Crucially, the spoofed speech signals in both sections have been generated using the same synthetic algorithms, minimizing other interference factors and facilitating precise measurement of the impact of language variation on detection accuracy. Table 5 provides an overview of the dataset composition.
- *Voice impersonation corpus in Hindi language (VIHL)*: This dataset consists of Hindi language dataset that contains 300 audios (150 bonafide audios and 150 spoofed audios that are created by applying impersonation attack). These audios have been collected from YouTube. Firstly, pre-processing has been done to convert the audio into frame. We have divided the audio into 3 s per frame. Table 6 gives the description about VIHL dataset. The

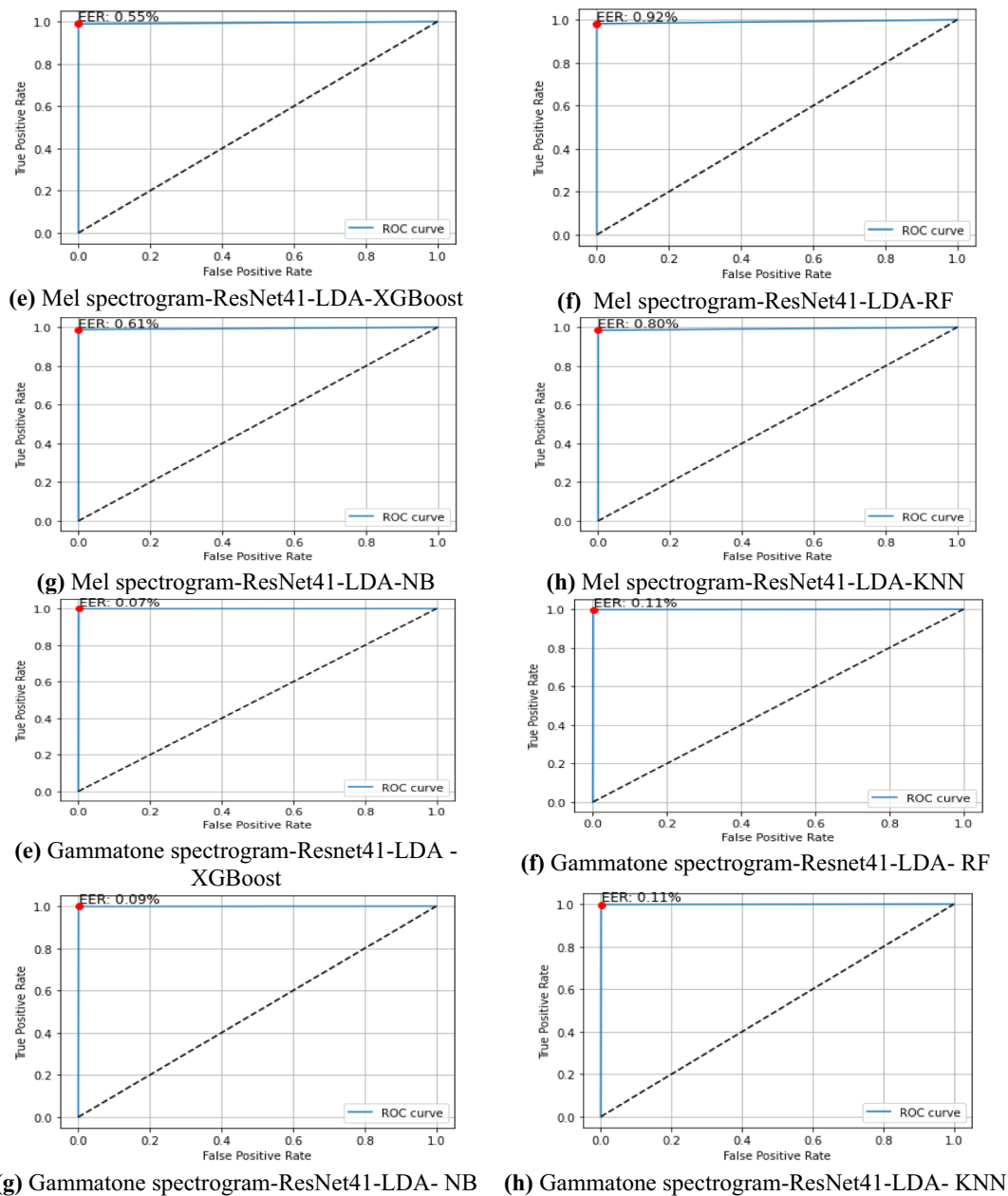


Fig. 12 DET Curve for Proposed Feature Combinations using DECRO Chinese

limitations of this dataset are that it contains very fewer audio samples for training and testing.

4.2 Performance analysis of baseline lfcc features with ML classifiers over different dataset

For challenge ASVspoof 2019, authors have given two baseline features for model development. Inspired from this, we

have used only one baseline feature that is LFCC with different machine learning models to check its performance over different datasets that are available publicly. From Table 7, it can be concluded that the baseline features LFCC with XGBoost give best result for datasets VSDC and DECRO English. However, give the worst performance on ASVs-poof 2019 and ASVspoof 2021 datasets containing audios of replay attack and deepfake attack, respectively.

Table 14 Performance of proposed feature combinations using VIHL

Frontend techniques	Classification Models	EER (%)	Accuracy	Recall	F1-Score	Precision
Mel spectrogram-ResNet41-LDA	XGBoost	2.50	98	97	100	95
	Random Forest	5.00	96	94	100	90
	Naïve Byes	2.50	98	97	100	95
	KNN	5.00	96	94	100	90
Gammatone spectrogram-ResNet41-LDA	XGBoost	0.92	99	99	100	99
	Random Forest	0.92	99	99	100	99
	Naïve Byes	0.92	99	99	100	99
	KNN	0.92	99	99	100	99

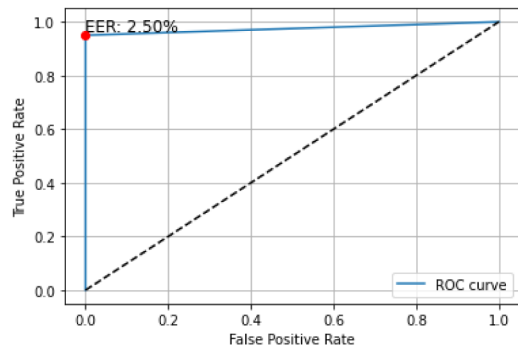
4.3 Performance analysis of proposed feature combinations with ML classifiers over different dataset

For the implementation of the proposed work, we have used two spectrograms, namely Mel spectrogram and Gammatone spectrogram. Both spectrograms have been given as input to ResNet41 for feature extraction individually, resulting in 1024 features for each audio sample. To streamline the frontend phase's complexity, dimensionality reduction has been performed using Linear Discriminant Analysis (LDA), yielding the most discriminative features. Subsequently, in the backend phase, the improved feature set has been subjected to four machine learning classifier, one by one, including XGBoost, RF, NB, and KNN. The performance evaluation has been done using ASVspoof 2019, 2021, VSDC, VIHL and DECRO datasets.

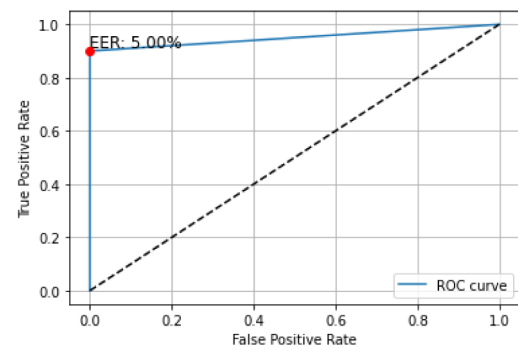
- *Performance of proposed feature combinations using ASVspoof 2019 LA:* Table 8 gives the results obtained for synthetic attack detection. At frontend two feature combinations Mel spectrogram-ResNet41-LDA and Gammatone spectrogram-LDA-ResNet41 have been used individually for extracting the relevant features from audios. The combinations of Melspect-LDA-ResNet41 with NB has achieved 5.81% EER and Gammaspect-LDA-ResNet41 with KNN has achieved EER of 1.88%. Also, the combination of Mel spectrogram-LDA-ResNet41 with XGBoost has achieved minimum EER 5.00% and the combination of Gammatone spectrogram-ResNet41-LDA with XGBoost has achieved minimum EER 1.76%. Also, it can be concluded that Gammatone spectrogram-ResNet41-LDA with XGBoost has outperformed all other proposed combinations. Figure 7 shows the Detection Error Trade-off (DET) curve for all the proposed combinations.
- *Performance of proposed feature combinations using ASVspoof 2019 PA:* Table 9 gives the results obtained for detecting replay attack. The combination of Mel spectro-

gram-LDA-ResNet41 with NB has achieved 12.44% and with XGBoost has achieved minimum EER 12.23%. The combination of Gammatone spectrogram-ResNet41-LDA has achieved second highest EER with NB by achieving 2.51% EER and with XGBoost has achieved minimum EER 1.28%. From results, it can be concluded that Gammatone spectrogram-ResNet41-LDA with XGBoost has outperformed other proposed combinations. Figure 8 shows the DET curve for all the proposed combinations.

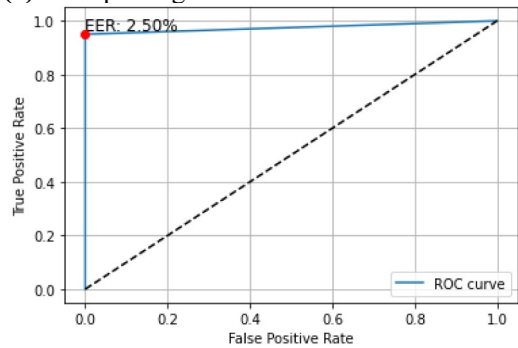
- *Performance of proposed feature combinations using ASVspoof 2021 DF:* Table 10 gives the results obtained for detecting deepfake attack. The combination of Mel spectrogram-ResNet41-LDA and Gammatone spectrogram-ResNet41-LDA has been applied, individually, for feature extraction from audio signal. For classification, four machine learning classifier RF, NB, KNN and XGBoost have been used, one by one. The combination of Mel-spectrogram-ResNet41-LDA with different classifiers have achieved same EER that is 5.00%. The combination of Gammatone spectrogram-ResNet41-LDA with Random Forest, Naïve Byes and KNN have achieved 0.92, 0.61, 0.80% respectively. However, combination of Gammatone spectrogram-ResNet41-LDA with XGBoost has achieved minimum EER 0.55%. From results, it can be concluded that Gammatone spectrogram-ResNet41-LDA with XGBoost has outperformed all other proposed combinations. Figure 9 shows the DET curve for all the proposed combinations.
- *Performance of proposed feature combinations using VSDC:* Table 11 gives the results obtained for detecting replay attack. VSDC dataset has been used for training and evaluating the proposed models. The combination of Mel spectrogram-ResNet41-LDA and Gammatone spectrogram-ResNet41-LDA has been applied, individually, for feature extraction from audio signal. For classification four machine learning classifier RF, NB, KNN and XGBoost have been used, one by one. The combination of Mel spectrogram-ResNet41-LDA with XGBoost, RF, NB, KNN have achieved minimum EER 0.56, 1.61, 0.70



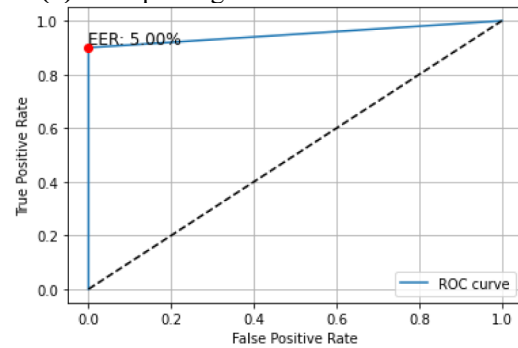
(a) Mel spectrogram-ResNet41-LDA- XGBoost



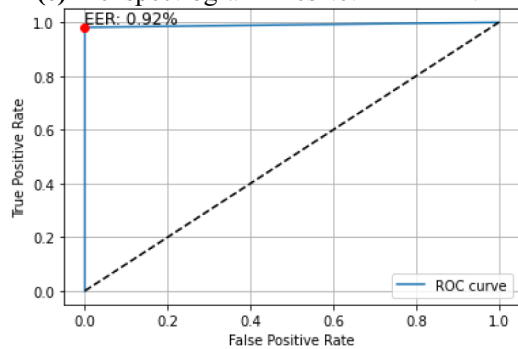
(b) Mel spectrogram-ResNet41-LDA- RF



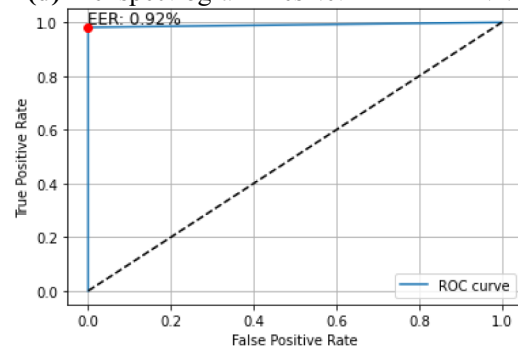
(c) Mel spectrogram- ResNet41-LDA-NB



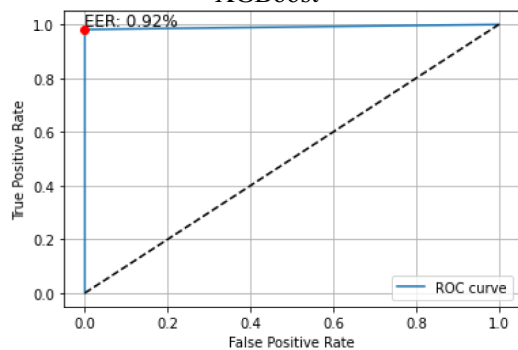
(d) Mel spectrogram-ResNet41-LDA- KNN



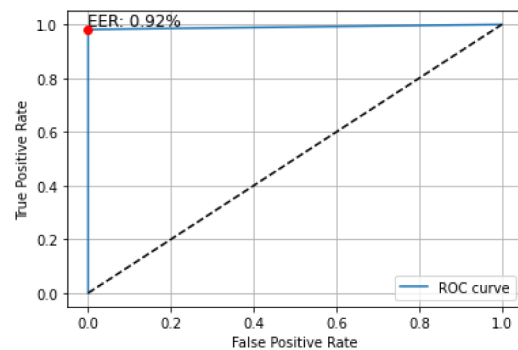
(e) Gammatone spectrogram-ResNet41-LDA- XGBoost



(f) Gammatone spectrogram ResNet41-LDA - RF



(g) Gammatone spectrogram-ResNet41-LDA- NB



(h) Gammatone spectrogram -LDA-Resnet41-KNN

Fig. 13 DET Curve for Proposed Feature Combinations using VIHL

Table 15 Comparison with existing methods for detecting different audio attacks

Methods	Dataset	EER%
<i>Synthetic Attack</i>		
MelSpectrogram-ResNet34(Aravind et al. 2020)	ASVspoof 2019 (LA-Eval)	2.98
CQCC-GMM(Yamagishi et al. 2019)		9.87
LFCC-GMM(Yamagishi et al. 2019)		11.96
FDLP + MFCC- GRU(Joshi and Dua 2023)		–
LFCC- XGBoost		15.5
Melspect + LDA + ResNet41-XGboost		5
Gammatone spectrogram-ResNet41-LDA -XGBoost		1.7
<i>Replay Attack</i>		
MelSpectrogram-ResNet34(Aravind et al. 2020)	ASVspoof 2019 (PA-Eval)	5.74
CQCC-GMM(Yamagishi et al. 2019)		11.02
LFCC-GMM(Yamagishi et al. 2019)		13.54
LFCC- XGBoost		21.1
Mel spectrogram-ResNet41-LDA-XGboost		12.2
Gammatone spectrogram-ResNet41-LDA -XGboost		1.28
FDLP + MFCC- GRU(Joshi and Dua 2023)	VSDC	1.6
LFCC- XGBoost		1.5
Mel spectrogram-ResNet41-LDA -XGboost		0.5
Gammatone spectrogram-ResNet41-LDA -XGboost		0.36
<i>Deepfake Attack</i>		
MelSpectrogram-ResNet34(Aravind et al. 2020)	ASVspoof 2021 (Deepfake-Eval)	–
CQCC-GMM(Yamagishi et al. 2019)		–
LFCC-GMM(Yamagishi et al. 2019)		–
FDLP + MFCC- GRU(Joshi and Dua 2023)		–
Pre-trained wav2vec2 + FIR-based data augmentation(Martín-Doñas and Álvarez 2022)		4.98
LFCC- XGBoost		40.6
Mel spectrogram-ResNet41-LDA -XGboost		5
Gammatone spectrogram-ResNet41-LDA -XGboost		0.5
Domain Adaptation + RawNet2 (Ba et al. 2023) (Chinese)	DECRO (English/ Chinese)	11.92
LFCC- XGBoost		2.8/6.87
Mel spectrogram-ResNet41-LDA -XGboost		1.28/0.5
Gammatone spectrogram-ResNet41-LDA -XGboost		0.03/0.07
<i>Mimicry Attack</i>		
Gammatone + ResNet27-XGBoost	VIHL	0.9
Mel spectrogram-ResNet41-LDA -XGboost		2.5
Gammatone spectrogram-ResNet41-LDA XGboost		0.9

and 1.28%, respectively. Also, the combination of Gammatone spectrogram-ResNet41-LDA with XGBoost has achieved minimum EER 0.36%, with RF 0.81%, with NB and KNN 0.46 and 0.61%, respectively. From results, it can be concluded that Gammatone spectrogram-ResNet41-LDA with XGBoost has outperformed other the proposed combinations. Figure 10 shows the DET curve for all proposed combinations.

- *Performance of proposed feature combinations using DECRO English:* Table 12 gives the results for detecting deepfake attack. The combinations of Mel spectrogram-ResNet41-LDA and Gammatone spectrogram-ResNet41-

LDA with Naïve Byes have achieved second minimum EER 1.88% and 0.05% EER. Also, the combinations of Mel spectrogram-ResNet41-LDA and Gammatone spectrogram-ResNet41-LDA with XGBoost have achieved minimum EER 1.28%, and 0.03%, respectively. From results, it can be concluded that Gammatone spectrogram-ResNet41-LDA with XGBoost has outperformed other proposed combinations. Figure 11 shows the DET curve for all the proposed combinations.

- *Performance of proposed feature combinations using DECRO Chinese:* Table 13 gives the results for detecting deepfake attack. The combinations of Mel spectrogram-

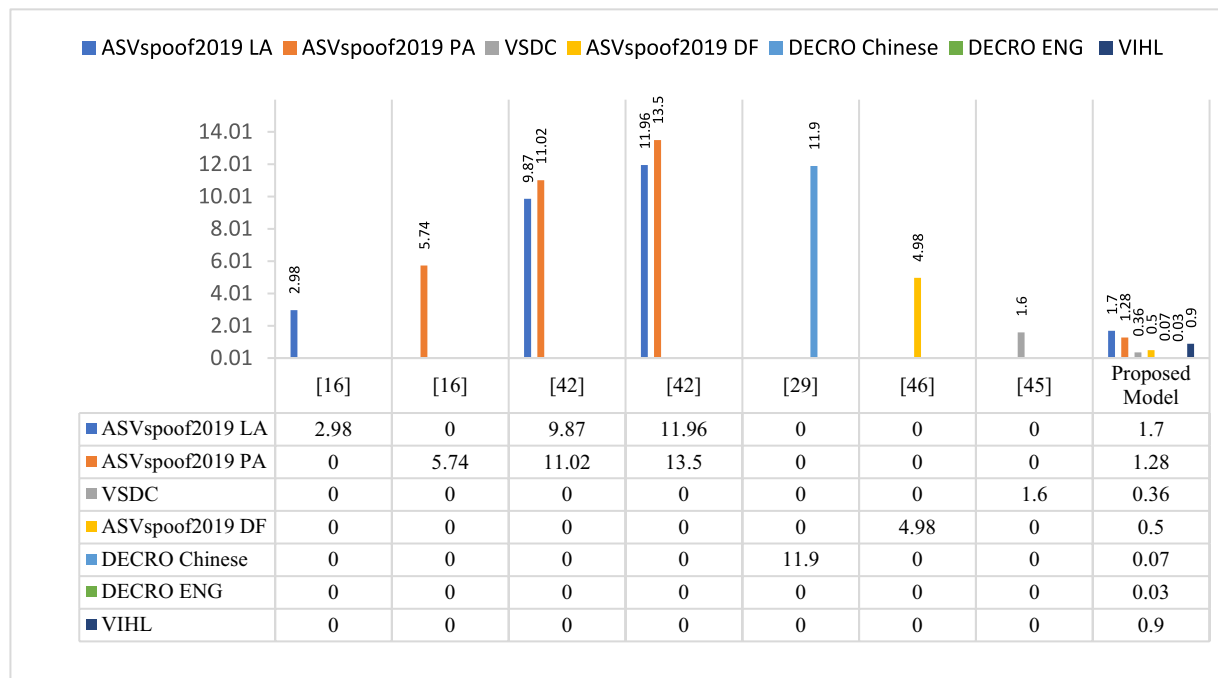


Fig. 14 Graphical Performance Comparison of Existing Countermeasures for ASD systems

ResNet41-LDA and Gammatone spectrogram-ResNet41-LDA with Naïve Byes have achieved second minimum EER 0.61 and 0.09% EER. Also, the combination of Mel spectrogram-ResNet41-LDA along with XGBoost has achieved minimum EER 0.55% and the combination of Gammatone spectrogram-ResNet41-LDA with XGBoost has achieved minimum EER of 0.07%. From results, it can be concluded that Gammatone spectrogram-ResNet41-LDA with XGBoost has outperformed the other proposed combinations. Figure 12 shows the DET curve for all proposed combinations.

- Performance of proposed feature combinations using VIHL:** Table 14 gives the results for detecting impersonation attack. VIHL dataset has been used for training and testing the proposed models. The combination of Mel spectrogram-ResNet41-LDA and Gammatone spectrogram-ResNet41-LDA has been applied, individually, for feature extraction from audio signal. For classification four machine learning classifier RF, NB, KNN and XGBoost have been used, one by one. The proposed combination Mel spectrogram-ResNet41-LDA with XGBoost, RF, NB, and KNN have achieved EER 2.50, 5.0, 2.5 and 5.00%, respectively. The combination Gammatone spectrogram-ResNet41-LDA with XGBoost, RF, NB, and KNN have achieved minimum EER 0.92, 0.92, 0.92 and 0.92%, respectively. From results, it can be concluded that Gammatone spectrogram-ResNet41-LDA with XGBoost has outperformed the other proposed combinations. Further,

it can be observed that the XGBoost classifier outperforms all the classifiers, while doing classification at the backend. Also, the combination of Gammatone spectrogram-ResNet41 is the best-performing model among all the implemented models, for detecting all four attacks using different datasets (Fig. 13).

5 Discussion and comparative analysis

This research aims to evaluate the effectiveness of different datasets in training and testing advanced algorithms for detecting manipulated audio. It involves using diverse publicly available datasets like ASVspoof 2019, ASVspoof 2021, VSDC, VIHL and DECRO, covering various spoofing scenarios in English, Hindi and Chinese.

The proposed model has been implemented in two parts. In the first part of the paper, a baseline for the proposed research work has been developed by comparing the performances of state-of-the-art baseline LFCC features with application of four different machine learning classifiers RF, KNN, XGBoost, and NB at the backend, over these four different datasets. In second part of the proposal, we have used novel feature combination of Melspect- ResNet41- LDA and Gammspect-ResNet41-LDA, one by one, with application of same set of machine learning classifiers at the backend.

From Table 7 it can be seen that combination of LFCC-XGBoost has achieved 15.5, 21.1, 37.1, 2.8, 6.06, 1.5, 5.8% using ASVspoof 2019 LA, ASVspoof 2019 PA, ASVspoof

2021 DF, DECRO English, DECRO Chinese, VSDC and VIHL datasets, respectively, for detecting synthetic, replay, deepfake and mimicry attack. From results it can be concluded that baseline worked well with DECRO, VSDC and VIHL dataset, however performed poorly on ASVspoof2019 and ASVspoof 2021 datasets.

Tables 8, 9, 10, 11, 12, 13, 14 shows the results of part two of proposed model. Table 8 shows the result of proposed combinations Mel spectrogram-ResNet41-LDA and Gammatone spectrogram-ResNet41-LDA with four machine learning classifiers. From Table 8, it can be seen that combination of Mel spectrogram-ResNet41-LDA with XGBoost and Gammatone spectrogram-ResNet41-LDA with XGBoost has produced 5 and 1.7% EER for detecting synthetic attack using ASVspoof2019 LA partition, respectively. From Table 9, it can be seen that combination of Mel spectrogram-ResNet41-LDA with XGBoost and Gammatone spectrogram-ResNet41-LDA with XGBoost has produced 12.2 and 1.28% EER for detecting replay attack using ASVspoof2019 PA partition, respectively. Using ASVspoof2021 dataset proposed combination of this work Mel spectrogram-ResNet41-LDA and Gammatone spectrogram-ResNet41-LDA with XGBoost have achieved 5% EER and 0.5% EER as given in Table 10. From Table 11, it can be deduced that combination of Mel spectrogram-ResNet41-LDA with XGBoost and Gammatone spectrogram-ResNet41-LDA with XGBoost has produced 0.15% and 0.36% EER for detecting replay attack using VSDC dataset. From Tables 12 and 13, it can be seen that combination of Mel spectrogram-ResNet41-LDA & Gammatone spectrogram-ResNet41-LDA with XGBoost have produced 1.28%, 0.03% EER and 0.55, 0.07% EER for detecting deepfake attack using DECRO English and Chinese dataset. Also to check the generalizability of our proposed model we have evaluated the model for Hindi language-based dataset. The proposed combinations Mel spectrogram-ResNet41-LDA & Gammatone spectrogram-ResNet41-LDA with XGBoost have produced 2.5 and 0.9% EER using VIHL dataset as given in Table 14.

From the results of Tables 12 and 13 it can be seen that proposed model performance degraded as languages changes from English to Chinese. Also, from the result discussed above, it can be concluded that language variation of same dataset can affect the results. In future, to solve the problem of language variability, distinct audio datasets with recordings in various languages and dialects are required. These datasets would help develop and test ASV models to ensure they work well in various language and cultural contexts.

From these results given by Tables 8, 9, 10, 11, 12, 13, 14, it can be concluded that combination of Gammatone spectrogram-ResNet41-LDA with XGBoost has outperformed other proposed combination for all the datasets. Table 15 shows the comparison between the suggested approach and existing methodology used for detecting different audio attacks.

Figure 14 shows the graphical comparison of existing countermeasures on the basis of EER (In Fig. 14 “0” represents Not Applicable).

6 Conclusion

The work proposed in this paper critically addresses the urgent need for robust security measures against audio spoofing amid the rapid advancements in voice-based technologies. The work thoroughly examined the performances of diverse datasets, such as ASVspoof challenge versions (2019, 2021), VSDC, VIHL and DECRO, encompassing various spoofing scenarios and languages, against baseline LFCC features, in its first part. The second part of the work applied Mel spectrogram-ResNet41-LDA and Gammatone spectrogram-ResNet41-LDA, one by one over publicly available dataset with application of four different machine learning classifiers at the backend. The proposed Gammatone spectrogram-ResNet41-LDA- XGBoost combination outperformed all the other combinations by exhibiting an EER of 1.7, 1.28, 0.5, 0.36, 0.03, 0.07, and 0.9% for ASVspoof 2019's Logical Access (LA), Physical Access (PA) partitions, ASVspoof 2021's Deepfake partition (DF), VSDC, DECRO English & Chinese, and VIHL dataset, respectively. The proposed approach has significant real-world applications, particularly in enhancing the security of voice authentication systems used in financial transactions, access control, and personal security. By improving the resilience of voice-based technologies against spoofing attacks, we believe that this research has the potential to increase trust and reliability in voice-based systems across various industries, making a substantial impact on safeguarding sensitive voice interactions.

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Data availability All data generated or analysed during this study are included in this published article.

Declarations

Conflict of interest All the authors declare no conflict of interest.

Ethical approval The data of this research do not involve any moral or ethical issues.

Informed consent All the authors are informed and approve the submission of this manuscript.

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